

Terrestrial broadcasting

The only form of transmission possible before the advent of the satellite Early Bird, in 1965, this form of transmission was limited in range by the curvature of the earth. Ranges were slightly over line of sight (due to some signals bouncing off the ionosphere, a layer of the atmosphere). Prior to the advent of communication satellites, the only way to transmit long distance, beyond visual range signals, was through a continuous system of high altitude aircraft maintained in a loop, a technique devised by Westinghouse radio. Today, terrestrial television is facing extinction due to strong competition from satellite television, a trend that might be reversed due to the recent introduction of high definition television (HDTV, explained later). HDTV can be transmitted only through terrestrial broadcasts due to the bandwidth limitations of satellite television. High definition television contains much more information in its images, and so requires a much larger bandwidth than standard television. Terrestrial television is received using a standard dipole antenna (the old ones in the shape of a skeleton). e.g. Early Doordarshan

Although terrestrial broadcasting has come nearly to an end, HD transmission may revive it unless satellites become capable of handling the high bandwidth

Satellite broadcasting

The predominant form of television broadcasting today, this began with the satellite Early Bird in 1965. Satellite broadcasting quickly gained popularity over terrestrial broadcasting due to the inherent cheapness of the technology. Terrestrial broadcasting to a geographically dispersed target audience was expensive, due to the large number of repeating stations and towers required. The first country to operate a nationwide fully operational satellite television network was the Soviet Union, through the Molniya series of satellites. Satellite broadcasted signals are received using a dish antenna and a set top box. The signals can be encrypted (coded in a way that can be decoded only by select set top boxes), making it possible to exact payment for subscription.